



PAK-MED STUDY CORNER

Your Success, Our Mission



TOTAL MCQs
100

GRAND TEST – 100 MCQs



BIOLOGY | CELL BIOLOGY

SECTION
01

1 The plant protoplast is surrounded by:

- A) Cell wall
- B) Cell membrane
- C) Tonoplast
- D) Plasma membrane

2 Nucleus is only visible when the cell is in:

- A) Dormant stage
- B) Maturation stage
- C) Dividing stage
- D) Non - dividing stage

3 Unlike eukaryotes, prokaryotes have:

- A) Nuclear membrane
- B) Ribosomes
- C) Circular chromosome
- D) Membranous organelles

4 The traffic of substances passing through nuclear membrane is controlled by definite structure of:

- A) Cisternae
- B) Nuclear envelope
- C) Nuclear pores
- D) Cristae

5 The plant cells differ from those of animals in lacking:

- A) Cell wall
- B) Asters
- C) Mitotic apparatus
- D) Microtubules

6 The Ribosomal RNA (rRNA) is synthesized and stored in:

- A) Mitochondria
- B) Nucleolus
- C) Lysosomes
- D) Ribosome

7 Animal and plant cells resemble in having:

- A) Plastids
- B) A covering around protoplast
- C) Centrioles
- D) Definite nucleus

8 Number of chromosomes found in sperms of chimpanzee are:

- A) 14 B) 16
- C) 24 D) 13

9 Central area of a prokaryotic cell is occupied by:

- A) Nucleoid
- B) Nucleus
- C) Sap surrounded by tonoplast
- D) Cytoplasm

10 Endoplasmic reticulum is often continuous with:

- A) Mitochondrial membrane
- B) Golgi apparatus
- C) Plasma membrane
- D) Thylakoid membrane

11 Prokaryotic cells carry out asexual reproduction by:

- A) Mitosis
- B) Binary fission
- C) Meiosis
- D) Budding

12 All of the following functions are performed by endoplasmic reticulum, EXCEPT:

- A) Protein synthesis
- B) Lipid metabolism
- C) Maintenance of cell shape
- D) Enzyme secretion

13 The rate of sedimentation of smaller subunits of the ribosomes of prokaryotes is:

- A) 30S
- B) 40S
- C) 50S
- D) 60S

14 Detoxification of harmful drugs in a cell is the function of:

- A) Mitochondrion
- B) Peroxisome
- C) RER
- D) SER

15 Movement of substances from lower concentration to higher concentration across semipermeable membrane is called:

- A) Downhill movement
- B) Passive transport
- C) Active transport
- D) Bulk transport



STUDY SMART
SCORE HIGH



FREE NOTES
PAST PAPERS



DAILY TESTS
MOCK TESTS



ACHIEVE YOUR
DREAM MBBS

16 The number of nuclear pores in differentiated cells such as erythrocytes have only:

- A) 10 to 20 pores/nucleus
- B) 5 or 10 pores/nucleus
- C) 1 to 2 pores/nucleus
- D) 3 or 4 pores/nucleus

17 Following processes can be carried out without involving membrane proteins, EXCEPT:

- A) Receptor-mediated endocytosis
- B) Pinocytosis
- C) Exocytosis
- D) Phagocytosis

18 Secretory granules (transport vesicles) are pinched off at the _____ face of Golgi apparatus:

- A) Forming face
- B) Maturing face
- C) Cis face
- D) Tran face

19 Unlike animals both prokaryotes and plants have:

- A) Cell membrane
- B) Cell wall
- C) Ribosomes
- D) Plasmids

20 The finishing, packing and distribution section of the cell is:

- A) ER
- B) Lysosome
- C) Peroxisome
- D) Golgi apparatus

21 The fluid located in the center of an animal cell is called:

- A) Cytoplasm
- B) Nucleoplasm
- C) Cell sap
- D) Cytosol

22 The presence of ribosomes and DNA indicates that mitochondria also synthesize some:

- A) Carbohydrates
- B) Lipids
- C) Proteins
- D) ATP

23 Previously the organisms were divided into:

- A) Prokaryotes and eukaryotes
- B) Animals and plants
- C) Bacteria and fungi
- D) Autotrophs and heterotrophs

24 Organelle involved in the synthesis of ATP is:

- A) Ribosome
- B) Mitochondrion
- C) Nucleus
- D) Centriole

25 A barrier between the cell contents and its environment is:

- A) Cell wall
- B) Cell membrane
- C) Nuclear membrane
- D) Tonoplast

26 _____ controls the life and activities of the cell:

- A) Nucleus
- B) Glyoxisome
- C) Ribosome
- D) Lysosome

27 Plant cell being an autotroph lacks:

- A) Photosynthesis
- B) Phagocytosis
- C) Centriole
- D) Cytokinesis

28 Disc-shaped protein structures within the centromere to which the spindle fibres are attach during mitosis or meiosis are:

- A) Kinetochores
- B) Centromeres
- C) Centrosomes
- D) Nucleosomes

29 Structure and chemical composition of prokaryotic _____ is different from that of eukaryotes:

- A) Plasma membrane
- B) Cell wall
- C) Ribosome
- D) Nuclear membrane

30 It is a complex system of interconnected tubules around the central stacks:

- A) Golgi apparatus
- B) Golgi complex
- C) Endoplasmic reticulum
- D) Centrioles

31 Both prokaryotic and eukaryotic cell walls do not differ with respect to:

- A) Composition
- B) Peptidoglycan
- C) Structure
- D) Location

32 The material of cell is recycled and renewed by an organelle called:

- A) Autophagosome
- B) Heterophagosome
- C) Primary lysosome
- D) Secondary lysosome

33 The carrier mediated, energy dependent movement of material into or out of cells, especially against concentration gradient is called:

- A) Passive transport
- B) Diffusion
- C) Active transport
- D) Osmosis

34 _____ is non-membranous, darkly stained body inside the nucleus:

- A) Ribosome
- B) Nucleolus
- C) Chromosome
- D) Nucleosome

35 According to the "Fluid Mosaic Model" of plasma membrane, inside the phospholipid bilayer, several types of _____ are embedded and protruding:

- A) Lipids
- B) Carbohydrates
- C) Proteins
- D) Nucleic acids

36 The proteins or enzymes which have to be transported out of the cell pass through the:

- A) Mitochondria
- B) Golgi apparatus
- C) Food vacuole
- D) SER

37 Unlike animal cells a plant cell is capable to maintain the composition of its cytoplasm and turgor pressure both, due to:

- A) Peripheral nucleus
- B) Central vacuole
- C) Cell wall
- D) Larger vacuole

38 Both plant and animal cells:

- A) Undergo osmotic lysis
- B) Carry out photosynthesis
- C) Carry out fermentation
- D) Carry out mitosis

39 Little bags inside the cells made up of phospholipids in which chemicals can be isolated from the external environment are known as:

- A) Organelles
- B) Vacuoles
- C) Lysosomes
- D) Mitochondria

40 Primitive type of cells lack following, EXCEPT:

- A) Chlorophyll
- B) Photosystem-I
- C) Discrete chromosomes
- D) Sap vacuoles

41 The phospholipid molecules keep on moving within their bilayer in a plasma membrane, is indicated by the term:

- A) Mosaic
- B) Fluid
- C) Unit membrane
- D) Bilayer

42 Pick up the one that is misfit among rest of the three choices:

- A) Passive transport
- B) Downhill movement
- C) Diffusion
- D) Active transport

43 _____ is also important for the mechanical stability of cell membrane:

- A) Cholesterol
- B) Sphingolipids
- C) Phospholipids
- D) Proteins

44 Maximum diversity with respect to the size of cell is found in:

- A) Prokaryotes
- B) Protists
- C) Eukaryotes
- D) Bacteria

45 Pick up the one not applicable to active transport:

- A) Movement in accordance with concentration gradient
- B) Movement against concentration gradient
- C) Carrier mediated transport
- D) Energy dependent movement

46 The phospholipid bilayer of plasma membrane is:

- A) Macroscopic structure
- B) Microscopic structure
- C) Sub microscopic structure
- D) Ultramicroscopic structure

47 Engulfment of bacteria by certain white blood cells is an example of:

- A) Exocytosis
- B) Endocytosis
- C) Pinocytosis
- D) Phagocytosis

48 The Okazaki fragments of prokaryotic cells are:

- A) 1000 – 2000 nucleotides long
- B) 100 – 200 nucleotides long
- C) 10000 – 20000 nucleotides long
- D) 10 – 20 nucleotides long

49 All prokaryotes including bacteria and cyanobacteria are:

- A) Single celled organisms
- B) Multicellular organisms
- C) Coenocytes
- D) Dikaryotic organisms

50 Any foreign object that gains entry into the cell is completely broken into simple digestible pieces by:

- A) Golgi apparatus
- B) Lysosomes
- C) Mitochondria
- D) Vacuoles

51 An automobile cylinder has a volume of 600 cm^3 at 1.0 atm and 7.0°C . When ignition occurs, the volume increases to 1200 cm^3 as the piston is moved and the pressure rises to 2.0 atm . What is the temperature of the cylinder after ignition occurs?

- A) 103°C
- B) 847°C
- C) 203°C
- D) 280°C

52 Gases deviate from ideal behavior at high pressure. Which of the following is correct statement for non-ideality?

- A) At high pressure, the collided gas molecules move in one direction only
- B) At high pressure, the collision between the gas molecules are increased manifold
- C) At high pressure, the volume of the gas becomes insignificant
- D) At high pressure, the intermolecular attractions become significant

53 Which of the following will have the same number of molecules at S.T.P?

- A) 44 g of CO_2 and 11.2 dm^3 of CO
- B) 28 g of N_2 and 5.6 dm^3 of oxygen
- C) 11.2 dm^3 of O_2 and 32 g of O_2
- D) 280 cm^3 of CO_2 and 280 cm^3 of N_2O

54 In the equation $\left(p + \frac{an^2}{V^2}\right)(V - nb) = RT$, (b) represents the _____:

- A) Excluded volume
- B) Excluded pressure
- C) Actual volume
- D) Excluded volume per mole

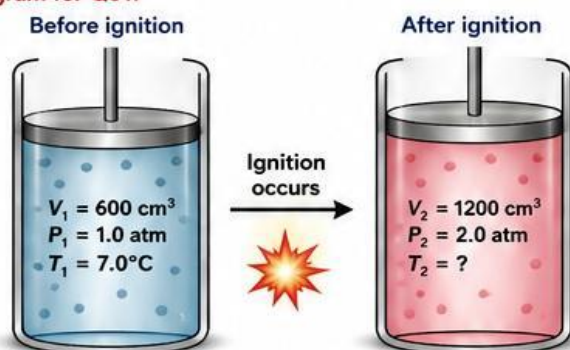
55 Under what condition van der Waal's gas equation for real gases is reduced to general gas equation:

- A) At high temperature and high pressure
- B) At high temperature and low pressure
- C) At low temperature and high pressure
- D) At low temperature and low pressure

56 According to Boyle's law at constant temperature, the product of pressure and volume of a given mass of gas is:

- A) Multiple
- B) A constant
- C) Fraction
- D) Whole number

Diagram for Q51:



57 Which of the following is SI unit of general gas constant (R)?

Opt.	Unit of P	Unit of V	Value of R
A)	atm	dm^3	$0.082 \text{ atm dm}^3 \text{ mol}^{-1} \text{ K}^{-1}$
B)	Nm^{-2}	cm^3	$82.05 \text{ atm cm}^3 \text{ mol}^{-1} \text{ K}^{-1}$
C)	Nm^{-2}	m^3	$8.3143 \text{ J mol}^{-1} \text{ K}^{-1}$
D)	Torr	dm^3	$62.4 \text{ torr dm}^3 \text{ mol}^{-1} \text{ K}^{-1}$

58 Molar volume of CO_2 is maximum at:

- A) STP
- B) 127°C and 1 atm
- C) 0°C and 2 atm
- D) 227°C and 2 atm

59 If pressure remains constant _____, to which temperature the volume of a gas will become twice to what it is at 0°C .

- A) 273 K
- B) 546 K
- C) 200°C
- D) 546°C

60 Which of the following gases shows minimum deviation from ideal gas behaviour?

- A) He
- B) H_2
- C) Ar
- D) O_2

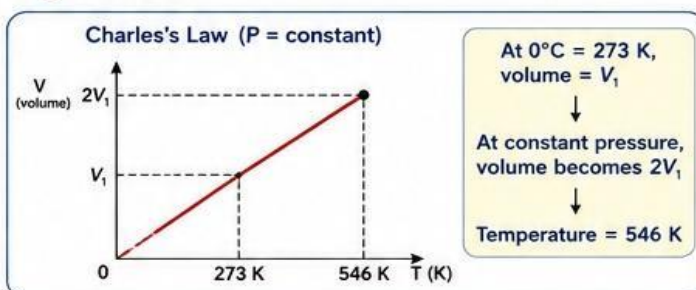
61 A real gas obeying van der Waal's equation will resemble ideal gas if:

- A) Both a and b are large
- B) a is small and b is large
- C) Both a and b are small
- D) a is large and b is small

62 According to Charles's law $\frac{V}{T} = k$ at constant pressure where k depends on all of the following factors EXCEPT:

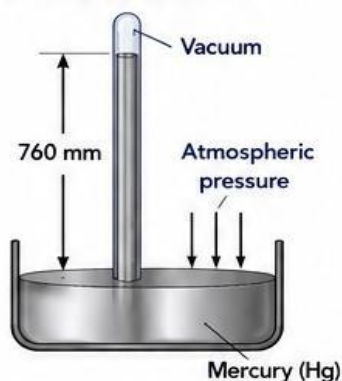
- A) Nature of the gas
- B) Mass of the gas
- C) Pressure of the gas
- D) Intramolecular forces

Diagram for Q59:



- 63** The pressure of air that can support 760 mmHg column at sea level is called one atmosphere (1 atm). It is the force exerted by 760mm long column of mercury of an area of 1cm^2 at 0°C . $1\text{ atm} = ?$

- A) 760 mm(Hg)
B) 14.5 PSI
C) 760 torr
D) 101325 Nm^{-2}

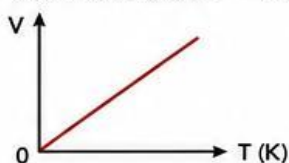


- 65** Which of the following diagram correctly describes the behavior of a fixed mass of an ideal gas (T is measured in kelvin)

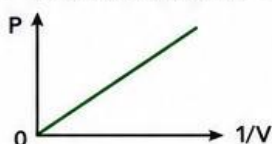
- A) P vs V at constant $T \rightarrow$ hyperbola



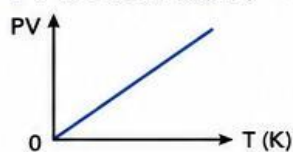
- B) V vs T at constant $P \rightarrow$ straight line through origin



- C) P vs $1/V$ at constant $T \rightarrow$ straight line

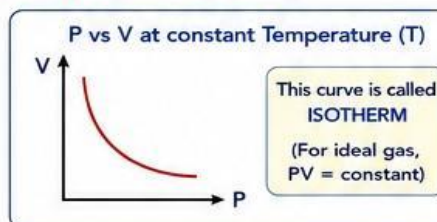


- D) PV vs T at constant $V \rightarrow$ straight line



- 68** When graph is plotted between pressure along x-axis and volume along y-axis at constant temperature, the curve obtained is called:

- A) Isotherm
B) Isobar
C) Isochor
D) Isotone



- 70** Calculate the density of NO_2 gas at 0°C and 1atm. pressure:

- A) $\frac{46 \times 1}{8.314 \times 300}\text{ gdm}^{-3}$
B) $\frac{46 \times 1}{0.0821 \times 273}\text{ gdm}^{-3}$
C) $\frac{46 \times 1}{1.985 \times 300}\text{ gdm}^{-3}$
D) $\frac{44 \times 1}{0.0821 \times 273}\text{ gdm}^{-3}$

Formula:
Density of gas,
 $d = \frac{PM}{RT}$

For NO_2 :
 $M = 46\text{ g mol}^{-1}$
 $P = 1\text{ atm}$
 $T = 273\text{ K } (0^\circ\text{C})$
 $R = 0.0821\text{ L atm mol}^{-1}\text{ K}^{-1}$

- 64** Which of the following laws can only be explained on the basis of Kelvin scale?

- A) Boyle's law
B) Charles's law
C) Dalton's law
D) Graham's law

- 66** Which of the following relationship is correct?

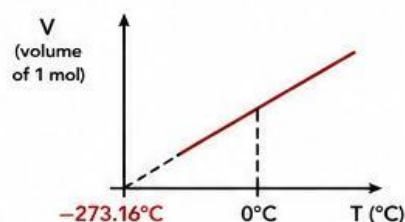
- A) $V \propto 1/T$
B) $V \propto d$
C) $V \propto P$
D) $n \propto 1/V$

- 67** If we plot a graph between temperature on x-axis and the volume of one mole of an ideal gas on y-axis, we get a straight line which cuts the temperature axis at -273.16°C .

This temperature is called absolute zero.

All of the following statements are correct about absolute zero EXCEPT:

- A) It is hypothetical temperature
B) It is unattainable
C) All the gases change into liquid state before this temperature
D) Absolute zero on different scales are 0K , -273.18°C and -450°F



At -273.16°C (0 K),
volume becomes zero
(ideal gas)

This temperature is
called **ABSOLUTE ZERO**.

- 69** Density of a gas is inversely proportional to:

- A) Molar mass of an ideal gas
B) Pressure on the gas
C) Temperature
D) Both A and B

For an ideal gas,

$$d = \frac{PM}{RT}$$

$$\text{Density (d)} \propto \frac{1}{M}, \frac{1}{P}, \frac{1}{T}$$

- 71** Which of the following chemical equations deals with the real gases?

- A) van der Waal's equation

$$\left(P_{\text{obs}} + \frac{an^2}{V^2}\right)(V_{\text{obs}} - nb) = nRT$$

- B) General gas equation $PV = nRT$

- C) Kinetic gas equation $PV = \frac{1}{3} mN \bar{C}^2$

- D) Root mean square velocity

$$C_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

van der Waals equation (option A) is used for **REAL GASES** as it takes into account intermolecular attractions (a) and volume of gas molecules (b).

72 Which one of the following expression represents the Avogadro law?

- A) $V = RnT/P$ (When T and n are constant)
- B) $V = RnT/P$ (When P and n are constant)
- C) $V = RnT/P$ (When P and T are constant)
- D) $V = RnT/P$ (When P , T and n constant)

73 Hydrogen gas possesses * __ * kinetic energy at the same temperature as compared to oxygen:

- A) More
- B) Same
- C) Less
- D) Sometimes less or sometimes more

74 Mark the incorrect statement:

- A) Barometer is used to measure atmospheric pressure
- B) Real gases are non-ideal gases
- C) At very low temperature and at very high pressure gases deviate from ideal gas
- D) SI unit of pressure is Nm^{-2} which is Pa

75 When 100 cm^3 of a gas at constant pressure is heated from 27°C to 100°C , the volume must be multiplied by?

- A) $\frac{373}{273}$
- B) $\frac{350}{300}$
- C) $\frac{300}{273}$
- D) $\frac{373}{100}$

76 How should the conditions be changed to prevent the volume of a given gas from expanding when its mass is increased?

- A) Temperature is lowered and pressure is increased
- B) Temperature is increased and pressure is lowered
- C) Temperature and pressure both are lowered
- D) Temperature and pressure both are increased

81 Which one of the following statements about states of matter is incorrect?

- A) The most abundant state around us is solid
- B) Less common state than solids, gases, and plasma is liquid
- C) Gases and liquids have three types of motions
- D) The most abundant gas in the universe is helium

77 Which of the following postulates of KMT explains Charles's law?

- A) Each gas consists of a large number of molecules
- B) The molecules of a gas have no forces of attraction
- C) The average K.E. of the gas molecules is directly proportional to absolute temperature
- D) The molecules of a gas are widely separated

78 Which of the following is the main causes of deviation of real gases from ideal behavior?

- A) No force of attraction among the gas molecules
- B) Molecules have negligible or practically no volume
- C) Both A and B
- D) Neither A and B

79 In a factory producing liquid air, one of the pipes carrying dry air -80°C is blocked with a white solid. What is this white solid?

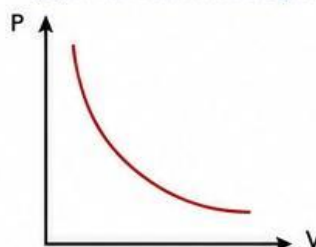
- A) Argon
- B) Ice
- C) Carbon dioxide
- D) Nitrogen

80 Identify the incorrect statement:

- A) $V_{\text{real}} = V_{\text{ideal}} - b$
- B) Van der waal's gas equation for one mole of a gas $\left(P + \frac{a}{V^2}\right)(V - b) = RT$
- C) $P_T = P_{\text{gas}} + P'$
- D) $P \propto \frac{1}{V}$

Diagram (Boyle's Law)

At constant temperature (T) for a fixed mass of gas

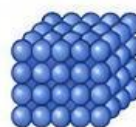


$$P \propto \frac{1}{V}$$

or

$$PV = \text{constant}$$

SOLID



Definite shape
Definite volume

LIQUID



No definite shape
Definite volume

GAS



No definite shape
No definite volume

82 If absolute temperature of a gas is doubled and pressure is reduced to half, the volume of the gas will:

- A) Remains unchanged
- B) Reduce to 1/4
- C) Increase four time
- D) Be doubled

83 Consider the following statements about kinetic gas equation ($PV = \frac{1}{3} mN \bar{C}^2$).

- I. N is number of molecules of gas in the vessel while N_A is 6.022×10^{23} number of gas molecules
- II. m is mass of one molecule of the gas while M_r is relative molecular mass of the gas
- III. $\bar{C}^2$ is mean square velocity of gas molecule while $\sqrt{\bar{C}^2}$ is root mean square velocity (rms)
- IV. $\bar{C}^2 = \frac{C_1^2 + C_2^2 + C_3^2 + \dots}{n_1 + n_2 + n_3 + \dots}$ while $C_{rms} = \sqrt{\frac{3RT}{M}}$

Which of the above statements is/are correct?

- A) I only
- B) II only
- C) III and IV
- D) I, II, III and IV

84 A gas has certain volume at 10°C . How much temperature should be raised to double its volume?

- A) 566 K
- B) 283 K
- C) 293 K
- D) 283°C

85 At the same temperature and pressure which of the following gases has the greatest density?

- A) CO_2
- B) SO_2
- C) Cl_2
- D) H_2O

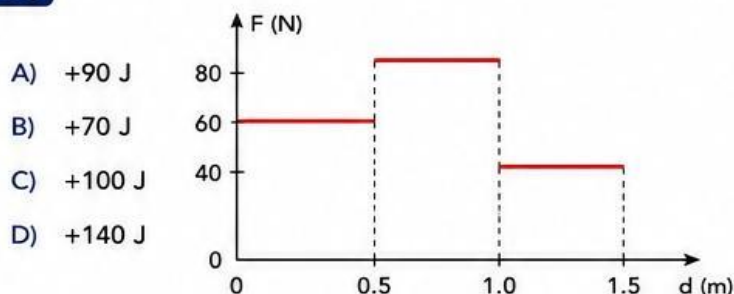
90 A force of $6\hat{i} + 12\hat{j} + 18\hat{k}$ N gives displacement of $2\hat{i} + 10\hat{j}$ m. The work done is:

- A) 132 J
- B) 140 J
- C) 144 J
- D) 156 J

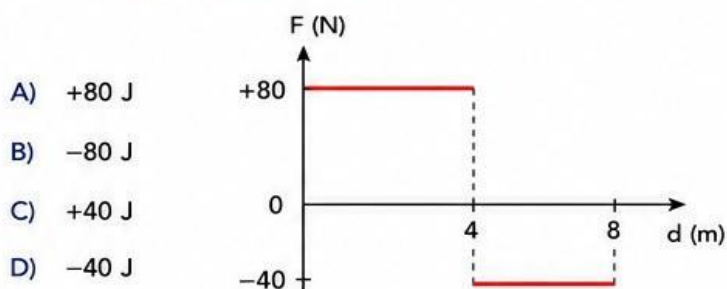
92 When a force does negative work on a moving car, then K.E of car?

- A) Increases
- B) Remains same
- C) Decreases
- D) May increase or decrease

86 The total work done for the given F-d graph is:



87 The work done for the given F-d graph up to displacement of 8 m is:



88 A body of mass 6 kg is under a force which causes displacement in it given by $d = \frac{t^2}{4}$ meters, where t is time. The work done by the force in 2 s is:

- A) 12 J
- B) 6 J
- C) 9 J
- D) 3 J

89 Two bodies X of mass 4 kg and Y of mass 6 kg have same linear momentum. If the K.E of Y is 48 J, then K.E of X is:

- A) 48 J
- B) 64 J
- C) 72 J
- D) 69 J

91 An object moves with uniform velocity of 40 m s^{-1} such that a force of 2000 N acts on it through a distance of 1.5 m by engine. What will be the work done by the frictional force if object remains in dynamic equilibrium?

- A) 0 J
- B) +3000 J
- C) -3000 J
- D) Can't be predicted

93 The engine of the car applies force of 250 N by which it moves with uniform velocity of 20 m s^{-1} . The power delivered by engine is:

- A) 5000 W
- B) 500 W
- C) 5 W
- D) 50,000 W

94

A car weighing 2 tons is moving half as fast as another car weighing 1 ton. The K.E. of the one ton car is:

- A) Less than that of the two-ton car
- B) Same as that of the two-ton car
- C) More than that of the two-ton car
- D) Impossible to compare

95

A wooden block has P.E of 400 J. When it is released from a certain height, its K.E at the bottom is 220 J. The work done against friction is:

- A) 200 J
- B) 180 J
- C) 220 J
- D) 150 J

96

If a body of mass 6 kg is dropped from the top of height 25 m, then its K.E at surface of earth is:

- A) 1300 J
- B) 1100 J
- C) 1700 J
- D) 1500 J

97

A man pulls a trolley through a distance of 60 m by applying a force of 70 N which makes an angle of 30° with horizontal. The work done by the man is:

- A) 870 J
- B) 4732 J
- C) 7266 J
- D) Zero

98

The area under the force vs displacement graph is called:

- A) Work
- B) Power
- C) Momentum
- D) Energy density

99

An engine pumps out 30 kg of water in one second. The water comes out vertically upwards with a velocity of 5 m s^{-1} , the power of engine in kilowatt is?

- A) 1.5
- B) 1500
- C) 15
- D) 150

100

If the mass of the body is made four times and the velocity becomes double, then the kinetic energy will increase:

- A) 16 times
- B) 12 times
- C) 24 times
- D) 18 times



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Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	21	B	41	B	61	B	81	B
2	B	22	B	42	B	62	B	82	B
3	B	23	B	43	B	63	B	83	B
4	D	24	B	44	B	64	B	84	B
5	B	25	B	45	B	65	B	85	B
6	B	26	B	46	B	66	B	86	B
7	B	27	B	47	B	67	B	87	B
8	B	28	B	48	B	68	B	88	B
9	B	29	B	49	B	69	B	89	B
10	B	30	B	50	B	70	B	90	B
11	B	31	B	51	B	71	B	91	B
12	B	32	B	52	B	72	D	92	B
13	B	33	B	53	B	73	B	93	B
14	B	34	B	54	B	74	B	94	B
15	B	35	B	55	B	75	B	95	B
16	B	36	B	56	B	76	B	96	B
17	B	37	B	57	B	77	B	97	B
18	B	38	B	58	B	78	B	98	B
19	B	39	B	59	B	79	B	99	B
20	B	40	B	60	B	80	B	100	B

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